THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE MCQ TEST SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 20 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) $\frac{20\sqrt{3}}{3}$ m **(b)** $20\sqrt{3}$ m **(c)** $10\sqrt{3}$ m **(d)** 10 m

2. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 60^\circ$. The $\angle OAB$ is: 1

(a) 90° **(b)** 30° **(c)** 60° **(d)** 120°

3. The radii of two circles are 6 cm and 8 cm. The radius of the circle whose area equals the sum of their areas is: 1

(a) 2 cm **(b)** 12 cm **(c)** 10 cm **(d)** 14 cm

4. From a solid cylinder of height 24 cm and base radius 7 cm, a conical cavity of the same height and the same base radius is hollowed out. The total surface area of the remaining solid is: 1

(a) 1760 cm² **(b)** 1606 cm² **(c)** 1364 cm² **(d)** 1914 cm²

5. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 3, 8, 20, 12, 7 respectively, The modal class is: 1

(a) 40–50 **(b)** 10–20 **(c)** 20–30 **(d)** 30–40

6. A bag contains 4 red and 6 blue balls. 2 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{7}{12}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{6}{13}$ **(d)** $\frac{5}{12}$

7. Which of the following is irrational? 1

(a) $\sqrt{25} - 3$ **(b)** $3 \times \sqrt{9}$ **(c)** $\sqrt{5} + 3$ **(d)** $\sqrt{9} + 3$

8. If α, β are zeroes of $x^2 - 9x + 14$, then $\alpha\beta$ is: 1

(a) 14 **(b)** 15 **(c)** 9 **(d)** -14

9. The pair $x + 2y = 3$ and $2x + ky = 7$ has no solution when $k =$ 1

(a) 6 **(b)** 5 **(c)** 4 **(d)** 2

10. The discriminant of $2x^2 - 7x + 3 = 0$ is: 1

(a) 25 **(b)** 37 **(c)** -25 **(d)** 73

11. The sum of first 6 positive odd numbers is: 1

(a) 72 **(b)** 42 **(c)** 30 **(d)** 36

12. In $\triangle ABC$, $AB = 9$, $BC = 12$, $AC = 15$. The triangle is: 1

(a) not right-angled **(b)** right-angled **(c)** equilateral **(d)** isosceles right

13. The distance of $(5, 3)$ from the y-axis is: 1

(a) 5 **(b)** 15 **(c)** 8 **(d)** 3

14. The value of $\sin 15^\circ / \cos 75^\circ$ is: 1

(a) 15 **(b)** 1 **(c)** 2 **(d)** 0

15. A storm breaks a tree at a height of 4 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 30° with the ground. The original height of the tree is: 1

(a) $12\sqrt{3}$ m **(b)** 8 m **(c)** 24 m **(d)** 12 m

16. A tangent to a circle of radius 9 cm has length 40 cm from the external point P to the point of contact. The distance of P from the centre of the circle is: 1

(a) 82 cm **(b)** 31 cm **(c)** 41 cm **(d)** 49 cm

17. The minute hand of a clock is 21 cm long. The area swept by the minute hand in 5 minutes is: 1

(a) 115.5 cm² **(b)** 11 cm² **(c)** 1386 cm² **(d)** 231 cm²

18. A wooden article is made by scooping out a hemisphere of radius 3.5 cm from each end of a solid cylinder of height 10 cm and base radius 3.5 cm. The total surface area of the article is: 1

(a) 451 cm² **(b)** 297 cm² **(c)** 220 cm² **(d)** 374 cm²

19. The mean of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 3, p, 20, 4, 7 respectively is 44.2. The value of p is: 1

(a) 16 **(b)** 15 **(c)** 21 **(d)** 18

20. A box contains 20 tickets numbered 1 to 20. One ticket is drawn at random. The probability that the number drawn is a multiple of 3 is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{7}{10}$ **(c)** $\frac{7}{20}$ **(d)** $\frac{3}{10}$

21. For which of the following is n such that 3^n ends with the digit 0? 1

(a) every n **(b)** $n = 2$ only

(c) only even n **(d)** no natural number n

22. The zeroes of $(x - 2)(x - 3)$ are: 1

(a) -2 and 3 **(b)** 2 and -3

(c) 2 and 3 **(d)** -2 and -3

23. The point where the lines $7x + 2y = -6$ and $3x + 5y = 14$ intersect has $x + y =$ 1

(a) 4 **(b)** -6 **(c)** 2 **(d)** -2

24. The nature of the roots of $2x^2 - 6x + 3 = 0$ is: 1

(a) three real roots **(b)** two distinct real roots

(c) no real roots **(d)** two equal real roots

25. The 10th term of the AP $\frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$ is: 1

(a) $\frac{9}{2}$ **(b)** 6 **(c)** 5 **(d)** $\frac{11}{2}$

26. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 2 : 3$. Then $AE : AC$ is 2: 1

(a) 4 **(b)** 5 **(c)** 6 **(d)** 3

27. The point $(x, 0)$ on the x-axis is equidistant from $(2, 3)$ and $(6, 1)$. Then $x =$ 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** -3

28. $(2\sin\theta)^2 + (2\cos\theta)^2 =$ 1

(a) 4 **(b)** 2 **(c)** 1 **(d)** 5

29. From the top of a building 7 m high, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. The height of the tower is: 1

(a) 14 m **(b)** $14 + 14\sqrt{3}$ m **(c)** $7\sqrt{3}$ m **(d)** $7 + 7\sqrt{3}$ m

30. In $\triangle ABC$, right-angled at B, $AB = 6$ cm and $BC = 8$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 2 cm **(b)** 3 cm **(c)** 5 cm **(d)** 4 cm

31. A horse is tied at a corner of a large square field with a rope 7 m long. The area of the field it can graze is: 1

(a) 77 m² **(b)** 11 m² **(c)** 154 m² **(d)** 38.5 m²

32. A solid metallic cylinder of base radius 6 cm and height 32 cm is melted and recast into solid spheres of radius 3 cm. The number of spheres obtained is: 1

(a) 32 **(b)** 38 **(c)** 64 **(d)** 30

33. The class mark of the class 10–25 is: 1

(a) 7.5 **(b)** 17.5 **(c)** 15 **(d)** 35

34. A lot of 135 ball pens contains 15 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{1}{9}$ **(b)** $\frac{8}{9}$ **(c)** $\frac{121}{135}$ **(d)** $\frac{119}{135}$

35. The exponent of 2 in the prime factorisation of 480 is: 1

(a) 5 **(b)** 7 **(c)** 8 **(d)** 6

36. The sum of the zeroes of the cubic $x^3 - 7x^2 + 14x - 8$ is: 1

(a) -7 **(b)** 7 **(c)** 8 **(d)** 14

37. The pair of equations $2x + 3y = 8$ and $4x + 6y = 12$ is: 1

(a) no solution (parallel lines)
(b) exactly two solutions
(c) exactly one solution (intersecting lines)
(d) infinitely many solutions (coincident lines)

38. For the equation $x^2 - 9x + 20 = 0$, the sum of the roots is: 1

(a) 20 **(b)** -9 **(c)** 9 **(d)** -11

39. The 15th term of $8, 13, 18, 23, \ldots$ is: 1

(a) 83 **(b)** 73 **(c)** 75 **(d)** 78

40. In $\triangle ABC$, $DE \parallel BC$, $AD = 3$ cm, $DB = 6$ cm, $AE = 6$ cm. Then EC (in cm) $=$ 1

(a) 15 **(b)** 12 **(c)** 6 **(d)** 18

41. The distance between $(-1, 2)$ and $(4, 14)$ is: 1

(a) 17 **(b)** 169 **(c)** 13 **(d)** 7

42. If $\sec\theta = \frac{5}{3}$ then $\cos\theta =$ 1

(a) $\frac{3}{5}$ **(b)** $\frac{5}{3}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{4}{3}$

43. A person looks at the top of a vertical pole from a point on the ground. The top of the pole is 10 m above the level of the person's eyes and the eyes are 10 m horizontally from the pole. The angle of elevation of the top of the pole is: 1

(a) 60° **(b)** 30° **(c)** 15° **(d)** 45°

44. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (3x + 5)$ cm and $QB = (2x + 9)$ cm. The length of QA is: 1

(a) 19 cm **(b)** 4 cm **(c)** 17 cm **(d)** 21 cm

45. The area of a sector of a circle of radius 21 cm is 231 cm^2. The central angle of the sector is: 1

(a) 90° **(b)** 60° **(c)** 30° **(d)** 120°

46. A solid metallic cone of base radius 6 cm and height 24 cm is melted and recast into a solid sphere. The radius of the sphere is: 1

(a) 12 cm **(b)** 5 cm **(c)** 6 cm **(d)** 7 cm

47. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50 with frequencies 8, 16, 8, 10, 8 respectively, The mean of the data is: 1

(a) 23.8 **(b)** 24.8 **(c)** 25 **(d)** 238

48. Two dice are thrown simultaneously. The probability that the sum of the numbers is 7 is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{5}{36}$ **(d)** $\frac{7}{36}$

49. The exponent of 5 in the prime factorisation of 3600 is: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

50. The number of zeroes of a polynomial whose graph cuts the x-axis at 2 point(s) is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 1

END OF QUESTION PAPER **Total: 50 Marks**

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